COURSE SLIDES

Gemini for Google Sheet (C817)

Course Code: C817

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome & Housekeeping

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About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

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YOUR TRAINER

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE
Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

CERTIFICATION
Industry certified in the subject area of this course.

DELIVERS
Professional short courses for Tertiary Infotech Academy.

FOUNDER
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

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ICE-BREAKER

Let's Know Each Other

1. Your name and organisation / role.

2. Your experience with this subject (if any).

3. What you want to be able to do after this course.

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HOUSEKEEPING

Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. Learn by doing — try every hands-on lab.

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COURSE PORTAL

Download Course Material

1. Sign in at the LMS-TMS portal
2. Open this course from your dashboard
3. Go to the Courseware tab
4. Download the Slides (PPT/PDF), Learner Guide and Lesson Plan
5. Keep them open alongside the labs as you work

Portal: https://lms-tms.tertiaryinfotech.com

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SCHEDULE

Lesson Plan — 1 Day, 8 hours/day

Day 1

- Day 1 — Getting started with Gemini AI in Google Sheets, then analysing and visualising real data with AI

– Topic 1: Getting Started with AI in Google Sheets (Labs 1–4)

– Topic 2: Analysing and Automating Data with AI in Sheets (Labs 5–8)

Day 1

- Day 1 — Getting started with Gemini AI in Google Sheets, then analysing and visualising real data with AI

– Topic 1: Getting Started with AI in Google Sheets (Labs 1–4)

– Topic 2: Analysing and Automating Data with AI in Sheets (Labs 5–8)

- Daily timing

– 9:30am–6:30pm · 1-hour lunch · tea breaks within training time

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WHAT YOU'LL ACHIEVE

Learning Outcomes

1

Understand & access AI

Explain what generative AI and Gemini in Google Sheets are, and set up and access AI features inside a sheet.

2

Generate & explain formulas

Generate spreadsheet formulas with AI and use AI to explain, step by step, what a formula does.

3

Prompt effectively

Write clear, effective prompts that get accurate results for spreadsheet tasks.

4

Clean & organise data

Use AI to clean, transform and organise messy spreadsheet data into a usable table.

5

Summarise & analyse

Summarise and analyse datasets with AI to surface useful, decision-ready insights.

6

Charts & visuals

Create tables, charts and visualisations from your data using AI.

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WHAT YOU'LL ACHIEVE

Learning Outcomes (cont.)

7

Validate & apply safely

Validate AI output for accuracy and apply AI safely and responsibly to real work data.

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LEARNING APPROACH

How You'll Learn

1. Trainer demonstrates the concept
2. You build it yourself in the hands-on lab
3. Verify with the lab's 'Test it' step
4. Discuss and troubleshoot together
5. Recap against the learning outcomes

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CORE CONCEPTS

Course Fundamentals

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OVERVIEW

How AI Works Inside Google Sheets

1. From typing formulas to describing results — you tell Gemini the outcome you want and it produces the formula or the data for you.

2. It works where your data lives — Gemini runs inside the sheet, using the ranges and tabs you already have.

3. Two ways in — the Gemini side panel for a conversation about your data, and the =AI() function for AI output right in a cell.

4. You stay accountable — AI drafts and suggests; you verify every result before it informs a decision.

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The Ask–Check–Apply Loop

1

Prepare

Know your data: which range, what each column means, and what a correct answer looks like.

2

Ask

Write a clear prompt — the range, the goal, the output format and any conditions.

3

Check

Verify the AI's formula or result against a value you can confirm yourself.

4

Apply

Put the verified result to work in your sheet.

5

Refine

Adjust the prompt and re-run until the output is right, then reuse it.

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THE WORKING RULE

AI in Sheets is fastest when your prompt is specific and your check is honest.

This course is hands-on: you clean, calculate, analyse and chart one real dataset with Gemini, verifying every result before you trust it.

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What You'll Build

A clean dataset

- A messy orders sheet standardised and organised
- Consistent dates, text and categories
- Formulas generated and explained by AI

An analysis

- Totals, averages and top performers
- A summary table / pivot
- A plain-language read of the numbers

A visual story, applied safely

- The right chart for the message
- AI outputs checked against known values
- Your own safe-use checklist

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THE LEARNING ARC

How Every Lab Works

1. The trainer demonstrates the AI technique on the shared Riverside Supplies dataset.
2. You do it yourself in Google Sheets — on the sample data, or on your own non-confidential data.
3. You verify the result against the lab's explicit 'Test it' check.
4. You compare the AI's output with a value you can confirm, and refine your prompt if it is off.
5. You keep the working prompt or formula — it becomes part of your AI-in-Sheets toolkit.

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01

TOPIC 01

Getting Started with AI in Google Sheets

Generative AI & Gemini in Sheets · Setting up and accessing AI · Generating and explaining formulas · Effective prompting

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Key Concepts — Getting Started with AI in Google Sheets

1. Generative AI — AI that creates new content (text, formulas, summaries) from a plain-language request, rather than only looking things up.

2. Gemini in Google Sheets — Google's built-in AI that helps you build formulas, clean data, analyse and summarise, directly inside a spreadsheet.

3. Where to find it — the Gemini side panel ("Ask Gemini") and AI-assisted menus let you work in plain English without leaving your sheet.

4. The =AI() function — a Sheets function that sends your prompt to Gemini and returns the result straight into a cell, so AI output becomes spreadsheet data.

5. Generating formulas — describe the calculation you want in words and let AI write the correct formula (SUM, IF, VLOOKUP, and more) for you.

6. Explaining formulas — paste any formula and ask AI to explain, step by step, what it does — a fast way to learn and to check an inherited sheet.

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Key Concepts — Getting Started with AI in Google Sheets (cont.)

7 Effective prompting — a good spreadsheet prompt names the range, the goal, the output format and any conditions, so the AI has no room to guess.

8 Human in the loop — AI drafts; you check. Always verify a formula or result against a value you can confirm before you rely on it.

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WHAT YOU'LL DO

Hands-On Labs — Getting Started with AI in Google Sheets

Labs 1–2

- Set Up and Access Gemini AI in Google Sheets
- Generate and Explain Formulas with AI

Labs 3–4

- Combine and Troubleshoot Formulas with AI
- Write Effective Prompts for Spreadsheet Tasks

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TOPIC 01 · HANDS-ON

Set Up and Access Gemini AI in Google Sheets

LAB 1

This lab gets AI working in your spreadsheet. You make your own copy of the Riverside Supplies orders sheet, open the Gemini side panel, and ask a first read-only question so you can see the ask-check loop before you rely on it. BUILDING BLOCK — what you do in this lab is applied to your copy of the Riverside Supplies sheet, the single dataset you clean, analyse and chart across all 8 labs.

You'll build

Your own copy of the Riverside Supplies sheet in Drive, with Gemini open and a first answer verified.

Tools: Google Sheets, Google Drive, Gemini side panel (Ask Gemini)

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LAB 1

Set Up and Access Gemini AI in Google Sheets

STEP 1 OF 8

1

Open the sample 'Riverside Supplies — Q1 Orders' sheet from the link the trainer shares.

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LAB 1

Set Up and Access Gemini AI in Google Sheets

STEP 2 OF 8

2

Make it yours: File > Make a copy, and save it to your Drive as 'Riverside Supplies — Q1 Orders (<YOUR NAME>)'.

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LAB 1

Set Up and Access Gemini AI in Google Sheets

STEP 3 OF 8

3

Look at the Orders tab: note the columns Order ID, Order Date, Customer, Region, Product, Category, Quantity, Unit Price, Total — and that some cells are messy. You will fix those later.

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LAB 1

Set Up and Access Gemini AI in Google Sheets

STEP 4 OF 8

4

Open the Gemini side panel — click 'Ask Gemini' (the sparkle, top-right). If you cannot see it, tell the trainer; access is confirmed at the start of the day.

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LAB 1

Set Up and Access Gemini AI in Google Sheets

STEP 5 OF 8

5

Ask a first, read-only question to confirm Gemini can see your data.

```
$ How many rows of orders are in the Orders tab, and what does each column mean?
```


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LAB 1

Set Up and Access Gemini AI in Google Sheets

STEP 6 OF 8

6

Check the answer: compare the row count Gemini gives against the row count shown at the bottom-left of the sheet.

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LAB 1

Set Up and Access Gemini AI in Google Sheets

STEP 7 OF 8

7

Ask one follow-up so you see Gemini use the data.

```
$ Which 3 products appear most often in the Orders tab?
```


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LAB 1

Set Up and Access Gemini AI in Google Sheets

STEP 8 OF 8

8

Confirm nothing changed — asking Gemini a question is read-only. You have now seen the ask-check loop safely.

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LAB 1 · VERIFY

Set Up and Access Gemini AI in Google Sheets

✅ Test it

You have your own copy of the sheet in Drive, the Gemini panel open, and a first answer whose row count matches the count shown at the bottom of the sheet.

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TOPIC 01 · HANDS-ON

Generate and Explain Formulas with AI

LAB 2

Now you let AI write formulas for you. You fill the blank Total column and add a grand total by asking Gemini for the formula, verify each against a value you can confirm, then paste an unfamiliar formula and have AI explain it step by step. BUILDING BLOCK — what you do in this lab is applied to your copy of the Riverside Supplies sheet, the single dataset you clean, analyse and chart across all 8 labs.

You'll build

A working, AI-generated Total column and grand total, each verified, plus a plain-English explanation of a formula.

Tools: Gemini side panel, formula generation, formula explanation, SUM, IF

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LAB 2

Generate and Explain Formulas with AI

STEP 1 OF 8

1

Click cell I2 (Total). Ask Gemini for a formula that multiplies Quantity by Unit Price.

```
$ Write a Google Sheets formula for cell I2 that multiplies Quantity in G2 by Unit Price in H2.
```


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LAB 2

Generate and Explain Formulas with AI

STEP 2 OF 8

2

Paste the formula into I2 and fill it down the column (double-click the small square at the cell's bottom-right).

```
$ =G2*H2
```


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LAB 2

Generate and Explain Formulas with AI

STEP 3 OF 8

3

Verify: pick any row and check by hand that Quantity x Unit Price equals the Total shown.

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LAB 2

Generate and Explain Formulas with AI

STEP 4 OF 8

4

Ask AI for one formula that totals the whole Total column into cell K1.

$ Write a formula for cell K1 that gives the grand total of the Total column I2:I200.

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LAB 2

Generate and Explain Formulas with AI

STEP 5 OF 8

5

Paste it into K1 and read the result.

```
$ =SUM(I2:I200)
```


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LAB 2

Generate and Explain Formulas with AI

STEP 6 OF 8

6

Cross-check: select I2:I200 and read the Sum shown in the status bar at the bottom-right — it should match K1.

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LAB 2

Generate and Explain Formulas with AI

STEP 7 OF 8

7

Learn from a formula: paste this into the Gemini panel and ask it to explain, step by step.

```
$ Explain what this formula does, step by step: =IF(G2>=10,"Bulk","Standard")
```


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LAB 2

Generate and Explain Formulas with AI

STEP 8 OF 8

8

Read the explanation and write one sentence in your notes on what =IF does.

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LAB 2 · VERIFY

Generate and Explain Formulas with AI

✅ Test it

The Total column is filled with a verified formula, K1 equals the status-bar Sum of I2:I200, and you can explain in one sentence what the =IF formula does.

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TOPIC 01 · HANDS-ON

Combine and Troubleshoot Formulas with AI

LAB 3

Real formulas need combining and fixing. You ask AI for a conditional total (sales for one region), hit the messy-data edge case that makes it wrong, and use Gemini to diagnose and correct it — verifying every version against a filtered check. BUILDING BLOCK — what you do in this lab is applied to your copy of the Riverside Supplies sheet, the single dataset you clean, analyse and chart across all 8 labs.

You'll build

A verified region-total formula (SUMIF) and an AI-assisted fix for a deliberately broken formula.

Tools: Gemini side panel, SUMIF, COUNTIF, troubleshooting, refinement

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LAB 3

Combine and Troubleshoot Formulas with AI

STEP 1 OF 8

1

In cell K3, ask AI for a formula that adds up Total for every order where Region is 'North'.

```
$ Write a formula for K3 that sums column I where column D equals "North", for rows 2 to 200.
```


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LAB 3

Combine and Troubleshoot Formulas with AI

STEP 2 OF 8

2

Paste it into K3 and note the figure.

```
$ =SUMIF(D2:D200,"North",I2:I200)
```


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LAB 3

Combine and Troubleshoot Formulas with AI

STEP 3 OF 8

3

Verify: filter the sheet to Region = North and read the status-bar Sum of the visible Total cells — it should match K3.

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LAB 3

Combine and Troubleshoot Formulas with AI

STEP 4 OF 8

4

Spot the problem: the Region column is messy ('north', 'North ', 'NORTH'), so SUMIF may miss rows and under-count. Tell AI and ask for a fix.

$ My Region column has inconsistent capitalisation and extra spaces, so SUMIF misses rows. Give me a formula that totals Total for North regardless of case or spaces.

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LAB 3

Combine and Troubleshoot Formulas with AI

STEP 5 OF 8

5

Paste the improved formula Gemini gives and compare the new North total with the old one.

```
$ =SUMIF(D2:D200,"*north*",I2:I200)
```


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LAB 3

Combine and Troubleshoot Formulas with AI

STEP 6 OF 8

6

Ask AI to explain, in one or two sentences, why the first formula missed rows.

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LAB 3

Combine and Troubleshoot Formulas with AI

STEP 7 OF 8

7

Break something on purpose: change K3 to reference the wrong column, then paste the wrong result to Gemini and ask what is wrong.

```
$ This formula gives the wrong total: =SUMIF(C2:C200,"North",I2:I200). What is wrong and how do I fix it?
```


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LAB 3

Combine and Troubleshoot Formulas with AI

STEP 8 OF 8

8

Apply Gemini's fix and confirm the corrected total matches your filtered check from step 3.

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LAB 3 · VERIFY

Combine and Troubleshoot Formulas with AI

✅ Test it

K3 gives a North total that matches your filtered status-bar Sum, and you have used AI to diagnose and correct a deliberately broken formula.

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TOPIC 01 · HANDS-ON

Write Effective Prompts for Spreadsheet Tasks

LAB 4

A result is only as good as the prompt. You run a vague prompt, then a specific one that names the range, goal, output and conditions, see the difference, and distil what worked into a reusable pattern you will use for the rest of the course. BUILDING BLOCK — what you do in this lab is applied to your copy of the Riverside Supplies sheet, the single dataset you clean, analyse and chart across all 8 labs.

You'll build

A written four-part prompt pattern and one strong, tested prompt saved for reuse.

Tools: Prompt design, range/goal/output/conditions, refinement

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LAB 4

Write Effective Prompts for Spreadsheet Tasks

STEP 1 OF 8

1

Run a deliberately vague prompt in the Gemini panel and note how generic the answer is.

```
$ Tell me about my sales.
```


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LAB 4

Write Effective Prompts for Spreadsheet Tasks

STEP 2 OF 8

2

Now run a specific prompt for the same intent and compare the result.

```
$ From the Orders tab range A1:I200, list the top 5 products by total sales value. Show product name and total, sorted highest first, as a table.
```


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LAB 4

Write Effective Prompts for Spreadsheet Tasks

STEP 3 OF 8

3

Write down the four parts that made the second prompt work: the Range, the Goal, the Output format, and the Conditions.

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LAB 4

Write Effective Prompts for Spreadsheet Tasks

STEP 4 OF 8

4

Capture your reusable pattern in a notes cell or a document so every future prompt follows it.

```
$ RANGE: <which cells/tab> | GOAL: <what to produce> | OUTPUT: <table/formula/number & where> | CONDITIONS: <filters, sort, limits>
```


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LAB 4

Write Effective Prompts for Spreadsheet Tasks

STEP 5 OF 8

5

Add conditions that keep results trustworthy: state the range explicitly and ask AI to show the formula it used so you can check it.

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LAB 4

Write Effective Prompts for Spreadsheet Tasks

STEP 6 OF 8

6

Rewrite one question of your own about the data using the pattern, and run it.

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LAB 4

Write Effective Prompts for Spreadsheet Tasks

STEP 7 OF 8

7

Refine once: change the Output or Conditions part (for example 'top 10', or 'North region only') and re-run to see the result change. Keep the better version.

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LAB 4

Write Effective Prompts for Spreadsheet Tasks

STEP 8 OF 8

8

Save your best prompt where you can find it again — you will reuse this pattern in every remaining lab.

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LAB 4 · VERIFY

Write Effective Prompts for Spreadsheet Tasks

✅ Test it

You can show two answers for the same intent (vague vs specific), a written four-part prompt pattern, and one refined prompt that produced the output you specified in the format you asked for.

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TOPIC RECAP

Recap — Getting Started with AI in Google Sheets

1. You can now: Open the sample sheet, make your own copy, open the Gemini side panel, and run a first AI prompt on the data.
2. You can now: Use AI to generate a spreadsheet formula from a plain-language description, and to explain an existing formula.
3. You can now: Use AI to build a multi-step formula, then diagnose and fix a broken one against a value you can confirm.
4. You can now: Compare a vague prompt with a specific one and capture a reusable four-part prompt pattern for spreadsheet work.

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02

TOPIC 02

Analysing and Automating Data with AI in Sheets

Cleaning, transforming & organising data · Summarising and analysing datasets · Tables, charts & visualisations · Validating and applying AI safely

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COURSE COVERAGE 50%

Key Concepts — Analysing and Automating Data with AI in Sheets

1. Cleaning data — AI can standardise text, fix inconsistent formats, split or merge columns and flag blanks, turning a messy sheet into a usable one.

2. Transforming data — reshape data with AI: extract fields, categorise rows, convert units or reformat dates without writing complex formulas by hand.

3. Organising data — sort, group and label records so a raw export becomes a structured table you can analyse.

4. Summarising datasets — ask AI for totals, averages, top and bottom performers and a plain-language summary of what the numbers say.

5. Analysing datasets — AI can spot trends, compare segments and answer "what" and "why" questions about your data in seconds.

6. Tables and pivots — turn a flat list into a summary table or pivot with AI, grouping and aggregating the figures that matter.

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COURSE COVERAGE 50%

Key Concepts — Analysing and Automating Data with AI in Sheets (cont.)

7 Charts and visualisations — describe the story you want to tell and let AI suggest and build the right chart to show it.

8 Validating and applying safely — AI can be confidently wrong; check outputs against known values, keep confidential data out of prompts, and stay accountable for the result.

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WHAT YOU'LL DO

Hands-On Labs — Analysing and Automating Data with AI in Sheets

Labs 5–6

- Clean, Transform and Organise Data with AI
- Summarise and Analyse Your Dataset with AI

Labs 7–8

- Create Tables, Charts and Visualisations with AI
- Validate AI Output and Apply AI Safely to Work Data

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TOPIC 02 · HANDS-ON

Clean, Transform and Organise Data with AI

LAB 5

Analysis is only as good as the data under it. You use Gemini to standardise the messy Region and Product text, fix the mixed date formats, and sort the records into a clean, consistent table — asking AI for the formula or steps each time and verifying the result. BUILDING BLOCK — what you do in this lab is applied to your copy of the Riverside Supplies sheet, the single dataset you clean, analyse and chart across all 8 labs.

You'll build

A cleaned, consistently formatted Orders table ready for analysis.

Tools: Gemini side panel, TRIM, PROPER, CLEAN, date formatting, Data > Sort range

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LAB 5

Clean, Transform and Organise Data with AI

STEP 1 OF 8

1

Ask AI how to standardise the Region column so 'north', 'North ' and 'NORTH' all become 'North'.

```
$ Give me a formula for column J that returns column D with consistent capitalisation and no leading or trailing spaces.
```


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LAB 5

Clean, Transform and Organise Data with AI

STEP 2 OF 8

2

Put the formula in J2 and fill it down; check that a few previously messy rows now read cleanly.

```
$ =PROPER(TRIM(D2))
```


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LAB 5

Clean, Transform and Organise Data with AI

STEP 3 OF 8

3

Lock the fix in: copy column J, then Edit > Paste special > Values only over column D, and delete the helper column J.

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LAB 5

Clean, Transform and Organise Data with AI

STEP 4 OF 8

4

Ask AI to standardise the Order Date column to YYYY-MM-DD, and follow its steps (Format > Number > Date, or the formula it gives).

$ Column B has dates in mixed formats. How do I make them all display as YYYY-MM-DD in Google Sheets?

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LAB 5

Clean, Transform and Organise Data with AI

STEP 5 OF 8

5

Verify a couple of dates against what they clearly should be, and confirm none turned blank or wrong.

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LAB 5

Clean, Transform and Organise Data with AI

STEP 6 OF 8

6

Tidy the Product column: ask AI for a formula to trim extra spaces and fix capitalisation, apply it, and paste values back as in step 3.

```
$ =TRIM(E2)
```


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LAB 5

Clean, Transform and Organise Data with AI

STEP 7 OF 8

7

Organise the data: select A1:I200 and use Data > Sort range (with 'Data has header row' ticked) to sort by Order Date, ascending.

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LAB 5

Clean, Transform and Organise Data with AI

STEP 8 OF 8

8

Ask Gemini to check your work.

$ Look at the Orders tab. Are there any remaining blank cells, inconsistent categories or duplicate Order IDs I should fix?

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LAB 5 · VERIFY

Clean, Transform and Organise Data with AI

✅ Test it

The Region and Product columns are consistent, all Order Dates display as YYYY-MM-DD, the data is sorted by date, and Gemini's check reports no remaining obvious issues (or you have fixed the ones it named).

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TOPIC 02 · HANDS-ON

Summarise and Analyse Your Dataset with AI

LAB 6

With clean data you can ask real questions. You have Gemini summarise the quarter, compare segments and describe the trend across the three months — and you verify every figure it gives against a formula or count you compute yourself before you trust it. BUILDING BLOCK — what you do in this lab is applied to your copy of the Riverside Supplies sheet, the single dataset you clean, analyse and chart across all 8 labs.

You'll build

A short, written analysis of the quarter in which every figure has been verified.

Tools: Gemini side panel, summarisation, AVERAGE, SUMIFS, segment analysis

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LAB 6

Summarise and Analyse Your Dataset with AI

STEP 1 OF 8

1

Ask AI for a summary of the quarter.

```
$ Summarise the Orders tab: total sales value, number of orders, average order value, and the best-selling product.
Show the figure for each.
```


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LAB 6

Summarise and Analyse Your Dataset with AI

STEP 2 OF 8

2

Verify the total sales against your grand total from Lab 2 (cell K1) — they should match.

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LAB 6

Summarise and Analyse Your Dataset with AI

STEP 3 OF 8

3

Verify the average order value yourself.

```
$ =AVERAGE(I2:I200)
```


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LAB 6

Summarise and Analyse Your Dataset with AI

STEP 4 OF 8

4

Ask a segment question.

$ Which Region had the highest total sales, and by how much more than the second-highest?

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LAB 6

Summarise and Analyse Your Dataset with AI

STEP 5 OF 8

5

Cross-check the winning region with a SUMIF like the one you built in Lab 3.

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LAB 6

Summarise and Analyse Your Dataset with AI

STEP 6 OF 8

6

Ask an analytical trend question.

$ Did sales grow or fall across the three months in this quarter? Give the monthly total for each month and describe the trend in one sentence.

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LAB 6

Summarise and Analyse Your Dataset with AI

STEP 7 OF 8

7

Verify the monthly totals with a quick pivot or a SUMIFS by month; if any figure does not match, investigate before trusting it.

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LAB 6

Summarise and Analyse Your Dataset with AI

STEP 8 OF 8

8

Write a 3-bullet summary of the quarter in a notes area, using only figures you have verified.

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LAB 6 · VERIFY

Summarise and Analyse Your Dataset with AI

✅ Test it

You have a written 3-bullet summary in which every figure — total sales, average order value and the top region — matches a formula or count you computed yourself.

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TOPIC 02 · HANDS-ON

Create Tables, Charts and Visualisations with AI

LAB 7

Numbers persuade when they are shown well. You use Gemini to design a pivot of sales by region and month, then to recommend and build the right chart, and to refine its title, labels and sorting — checking each visual against the figures you verified in Lab 6. BUILDING BLOCK — what you do in this lab is applied to your copy of the Riverside Supplies sheet, the single dataset you clean, analyse and chart across all 8 labs.

You'll build

A verified pivot table plus a labelled column chart and a monthly-trend line chart.

Tools: Gemini side panel, pivot tables, Insert > Chart, chart types, chart formatting

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LAB 7

Create Tables, Charts and Visualisations with AI

STEP 1 OF 8

1

Ask AI which summary table best shows sales performance and how to build it.

$ I want to see total sales by Region and by month. What pivot table should I build, and what are the steps in Google Sheets?

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LAB 7

Create Tables, Charts and Visualisations with AI

STEP 2 OF 8

2

Build the pivot (Insert > Pivot table): Rows = Region, Columns = month of Order Date, Values = SUM of Total.

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LAB 7

Create Tables, Charts and Visualisations with AI

STEP 3 OF 8

3

Verify one cell of the pivot against a SUMIFS you write for that same region and month.

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LAB 7

Create Tables, Charts and Visualisations with AI

STEP 4 OF 8

4

Ask AI to recommend the best chart to compare sales across regions.

```
$ What is the best chart type to compare total sales across Regions, and why?
```


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LAB 7

Create Tables, Charts and Visualisations with AI

STEP 5 OF 8

5

Insert the chart: select the region totals, then Insert > Chart, and set the recommended type (for example a column chart).

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LAB 7

Create Tables, Charts and Visualisations with AI

STEP 6 OF 8

6

Ask AI to improve it, then apply the suggestions — a clear title, axis labels, and bars sorted highest to lowest.

```
$ Suggest a clear title and axis labels for a column chart of total sales by Region, sorted highest to lowest.
```


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LAB 7

Create Tables, Charts and Visualisations with AI

STEP 7 OF 8

7

Add a second chart for the monthly trend: a line chart of total sales by month.

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LAB 7

Create Tables, Charts and Visualisations with AI

STEP 8 OF 8

8

Check the trend chart matches the direction you found in your Lab 6 analysis.

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LAB 7 · VERIFY

Create Tables, Charts and Visualisations with AI

✅ Test it

You have a pivot table whose figures you verified against a formula, a labelled column chart of sales by region sorted high-to-low, and a line chart of the monthly trend consistent with your Lab 6 analysis.

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TOPIC 02 · HANDS-ON

Validate AI Output and Apply AI Safely to Work Data

LAB 8

The capstone. You deliberately test whether AI's answers hold up, learn to verify anything before you act on it, review what data is safe to put into a prompt, and write a safe-use checklist you apply to an earlier result — the habits that make AI in Sheets trustworthy at work. BUILDING BLOCK — what you do in this lab is applied to your copy of the Riverside Supplies sheet, the single dataset you clean, analyse and chart across all 8 labs.

You'll build

A verification habit and a written safe-use checklist you have applied to an earlier lab's result.

Tools: Validation, cross-checking, data privacy, safe-use checklist

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LAB 8

Validate AI Output and Apply AI Safely to Work Data

STEP 1 OF 8

1

Ask AI a question with a knowable answer, then verify it with a formula.

```
$ How many orders are from the North region?
```


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LAB 8

Validate AI Output and Apply AI Safely to Work Data

STEP 2 OF 8

2

Confirm the count yourself and compare.

```
$ =COUNTIF(D2:D200,"North")
```


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LAB 8

Validate AI Output and Apply AI Safely to Work Data

STEP 3 OF 8

3

Try to catch a mistake: ask a trickier question and verify hard by recomputing it yourself.

$ What percentage of total sales came from the top 3 customers? Show the range you used and the working.

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LAB 8

Validate AI Output and Apply AI Safely to Work Data

STEP 4 OF 8

4

If the AI figure is wrong or unclear, refine the prompt (state the range and ask it to show its working) and re-run.

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LAB 8

Validate AI Output and Apply AI Safely to Work Data

STEP 5 OF 8

5

Set the rule: never accept an AI number you cannot tie back to a formula or count in the sheet.

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LAB 8

Validate AI Output and Apply AI Safely to Work Data

STEP 6 OF 8

6

Review data safety: list which columns here would be sensitive if this were real company data (for example Customer names), and how you would protect them.

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LAB 8

Validate AI Output and Apply AI Safely to Work Data

STEP 7 OF 8

7

Ask AI for good practice, then sanity-check its advice against your own judgement.

```
$ What should I avoid putting into an AI prompt when working with real company data, and why?
```


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LAB 8

Validate AI Output and Apply AI Safely to Work Data

STEP 8 OF 8

8

Draft your safe-use checklist — verify every figure; keep confidential data out of prompts; state the range; ask for the formula; record where AI was used — and apply it to one AI result from an earlier lab.

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LAB 8 · VERIFY

Validate AI Output and Apply AI Safely to Work Data

✅ Test it

You have caught or ruled out at least one AI inaccuracy by verifying it against your own formula, and you have a written safe-use checklist that you applied to an earlier lab's result.

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TOPIC RECAP

Recap — Analysing and Automating Data with AI in Sheets

1. You can now: Use AI to standardise messy text and dates, tidy fields, and organise the dataset into a clean table.

2. You can now: Use AI to summarise the dataset and answer analytical questions, verifying every figure against a formula.

3. You can now: Use AI to build a summary/pivot table and the right, clearly labelled chart to tell the data's story.

4. You can now: Stress-test AI output for accuracy and write a personal checklist for applying AI safely to real work data.

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WRAP-UP

Course Summary & Next Steps

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LEARNING OUTCOMES

What You Achieved

1

Understand & access AI

Explain what generative AI and Gemini in Google Sheets are, and set up and access AI features inside a sheet.

2

Generate & explain formulas

Generate spreadsheet formulas with AI and use AI to explain, step by step, what a formula does.

3

Prompt effectively

Write clear, effective prompts that get accurate results for spreadsheet tasks.

4

Clean & organise data

Use AI to clean, transform and organise messy spreadsheet data into a usable table.

5

Summarise & analyse

Summarise and analyse datasets with AI to surface useful, decision-ready insights.

6

Charts & visuals

Create tables, charts and visualisations from your data using AI.

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LEARNING OUTCOMES

What You Achieved (cont.)

7

Validate & apply safely

Validate AI output for accuracy and apply AI safely and responsibly to real work data.

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Keep Practising After the Course

1. Re-run the labs on your own machine — repetition is what makes the skills stick.
2. Apply the techniques to a real process or project in your own organisation.
3. Your course material stays available at https://lms-tms.tertiaryinfotech.com/.
4. Explore the follow-on courses at www.tertiarycourses.com.sg.

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THANK YOU FOR ATTENDING

Thank You!

You now have the practical skills from every lab in this course — keep applying them to your own work.

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